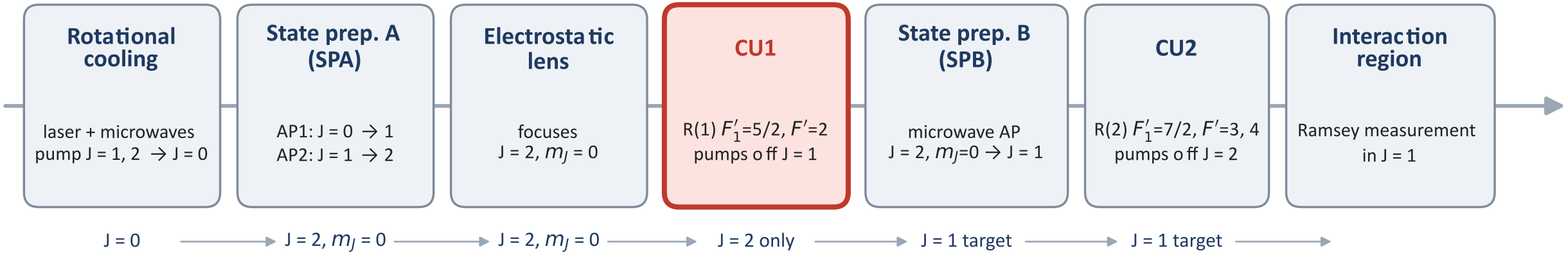
Goal of the cleanup: pump off J = 1 before SPB (CU1) and J = 2 after SPB (CU2)



state after each stage

Why CU1?

- J = 1 molecules still reach SPB: thermal J = 1 left by rotational cooling, and J = 1, m_J = 0 left by an incomplete AP2 (J = 1 → 2).
- In SPB the adiabatic passage would transfer them back, J = 1 → J = 2, or pass to the interaction region mixed with the prepared molecules

  → extra noise and systematics downstream. CU1 removes all of J = 1 before SPB.